SIREN: Detecting When a Sensor Is Quietly Failing, Without Being Told

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Abstract

Systems that combine several sensors—camera, microphone, motion sensors, text—usually assume every input is trustworthy, or at least that they will be *told* when one is missing. Reality is harsher: a sensor often keeps working on the surface while its data has become meaningless. A camera stays on but the room is dark; a microphone records only background hum; a motion sensor freezes at a fixed value after a glitch. The data still arrives and looks normal, so the system trusts it and produces a confident but wrong answer. We call this *silent sensor failure* and present **SIREN**, a method that detects it on its own, without anyone marking which sensor went bad. The key idea is simple: judge each sensor in three independent ways—does it *agree with the other sensors*, does it *agree with its own recent past*, and is its signal *believable on its own* and trust it only if it passes all three. A separate watcher detects whole-system breakdowns, and when the evidence is hopeless the system *abstains* instead of guessing. Using illustrative, clearly-labeled simulations, we show the intended behaviour: SIREN keeps working when one or more sensors fail quietly and signals low confidence exactly when it should.

1 Introduction

Multimodal AI is now everywhere: assistive robots, health monitors, driver-assistance, industrial inspection. These systems are strong when their inputs are clean, but real sensors fail all the time—and the *quiet* failures are the dangerous ones. A clearly *missing* sensor is manageable, because the system knows it is gone. A sensor that keeps streaming *plausible-looking but meaningless* data is much worse: nothing warns the system, so it blends the bad signal with the good and answers with full confidence.

Most existing robustness methods sidestep this. Most robustness methods assume missing inputs are explicitly flagged, but real-world systems must handle unflagged, silently failing sensors.

This paper takes silent failure as the problem to solve, and aims for a method that is easy to understand and to implement. Our contributions are:

- **A clear problem:** detect quietly failing sensors with no "missing" flag and no labels saying which sensor failed.
- **Three independent reliability checks** per sensor (agreement with others, agreement with its own past, standalone plausibility), combined with a simple "trust only if it passes all three" rule, so no single check is a weak point.

- **Safe behaviour:** a whole-system collapse detector and an *abstain* option, so the system says "I am not sure" instead of giving a confident wrong answer.

2 Related Work

Robust multimodal learning has mostly studied the *missing-modality* setting, where the absent input is known and is reconstructed or down-weighted [Ma et al., 2021, Zhao et al., 2021, Wang et al., 2024]. Quality-aware fusion methods weight each input by an estimated quality or reliability [Han et al., 2021, Zhang et al., 2023], which is the closest prior idea to ours, but they are trained with task labels and assume the inputs are genuine rather than silently corrupted. Our temporal check is a reliability use of predictive self-supervision [van den Oord et al., 2018]; our plausibility check is a form of density / out-of-distribution scoring [Hendrycks & Dietterich, 2019]. SIREN differs by (i) not requiring a missing-flag, (ii) combining *three independent* reliability signals so that correlated failures are harder to hide, and (iii) abstaining on collapse.

3 The Idea in One Picture

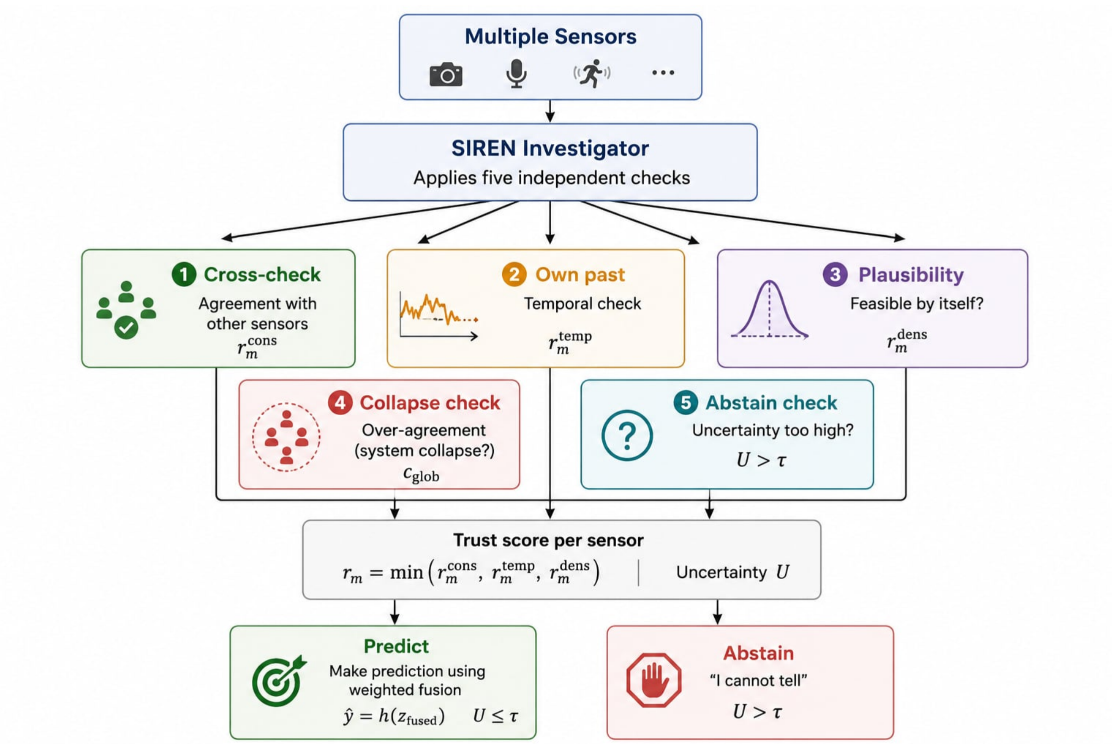

Figure 1: Overview of the proposed SIREN framework.

Intuition behind the three checks. Think of several witnesses describing the same event, where some may be unreliable without realizing it. A careful investigator would not simply trust the most confident witness. Instead, they would:

1. cross-check the stories against each other;
2. check each witness against what they said a moment earlier;

3. ask whether each story is even possible on its own;
4. become suspicious if everyone suddenly gives a suspiciously perfect, identical story;
5. say "I cannot tell" when the evidence is hopeless.

SIREN plays the role of this investigator, applied to sensors. In this analogy, checks (1)–(3) correspond to the three reliability signals, check (4) is the collapse detector, and check (5) corresponds to abstention.

Why three checks? The natural idea—"does this sensor agree with the others?"—fails when multiple sensors break simultaneously (for example, darkness disables both the camera and the activity-driven audio sensor). In that case, the broken sensors may still agree with each other and appear valid.

For this reason, checks (2) and (3) are essential: they do not depend on other sensors, so they remain informative even when several modalities fail together.

4 Method

4.1 Setup and notation

We have M sensors. Each sensor m is read over a short time window and passed through its own encoder f_m (a Transformer for text/audio, a vision model for video, a small 1D-CNN for numeric sensor streams). The encoder outputs a short *sequence* of feature vectors and a single *pooled summary*:

$$(z_m^1, \dots, z_m^T) = f_m(x_m), \qquad \bar{z}_m = \tfrac{1}{T} \sum_{t=1}^{T} z_m^t, \tag{1}$$

where each vector is L_2-normalized and has the same size d for every sensor.

Every sensor's input becomes (a) a short series of feature vectors over time and (b) one average "summary" vector. Using the same size for all sensors lets us compare them directly. The time series is used by the temporal check; the summary is used by the other checks and by fusion.

4.2 Module 1 — Consensus check (agreement with the other sensors)

A small predictor R tries to guess one sensor's summary from another's. The guessing error measures how consistent two sensors are:

$$e_{m \leftarrow j} = \|\bar{z}_m - R(\bar{z}_j)\|^2, \qquad s_{m \leftarrow j} = \frac{e_{m \leftarrow j} - \mu_{mj}}{\sigma_{mj}}, \tag{2}$$

where μ_{mj}, σ_{mj} are the average and spread of this error measured on clean data. The consensus score is

$$r_m^{\text{cons}} = \sigma\Big(- \beta \cdot \text{median}_{j \neq m}\, s_{m \leftarrow j} \Big) \in [0, 1]. \tag{3}$$

We check how badly each *other* sensor fails to predict sensor m, compare that error to what is normal for that pair, and take the median across all other sensors. If sensor m disagrees with the others more than usual, its score is low. The median means one rogue sensor cannot swing the result.

4.3 Module 2 — Temporal check (agreement with its own past)

This check uses only sensor m itself. A small predictor T_m guesses the next feature vector from the previous ones; the average prediction error is the "innovation":

$$a_m = \tfrac{1}{T-k} \sum_{t=k+1}^{T} \left\| z_m^t - T_m(z_m^{t-k:t-1}) \right\|^2, \qquad \tilde{a}_m = \frac{a_m - \mu_m^T}{\sigma_m^T}, \tag{4}$$

$$r_m^{\text{temp}} = \exp\!\big(- \tfrac{1}{2}\, \tilde{a}_m^2 \big) \in (0, 1]. \tag{5}$$

We ask each sensor: does what you say now follow naturally from what you were saying a moment ago? We compare its change-over-time to what is normal (again, learned on clean data). The bell-shaped score is high when the change is *typical*, and low in both bad cases: a frozen sensor changes *too little* and an erratic/noisy sensor changes *too much*. Crucially, this needs no other sensor, so it still works when several sensors fail together.

4.4 Module 3 — Plausibility check (believable on its own)

On clean data we fit a simple model p of what normal summary vectors look like for each sensor. The plausibility score is the (standardized) likelihood:

$$r_m^{\text{dens}} = \sigma\!\Big(\frac{\log p(\bar{z}_m) - \mu_m^L}{\sigma_m^L} \Big) \in [0, 1]. \tag{6}$$

Ignoring the other sensors and the past, does this reading even look like a normal reading for this sensor? A stuck-at-zero or pure-noise summary lands in an unusual region and scores low. This also helps tell "broken and frozen" apart from "genuinely quiet but fine," because a valid quiet reading still looks normal.

4.5 Combining the three checks

A sensor is trusted only if it passes *all three* checks:

$$r_m = \min\big(r_m^{\text{cons}},\ r_m^{\text{temp}},\ r_m^{\text{dens}} \big) \in [0, 1]. \tag{7}$$

We take the smallest of the three scores. So any single check can raise the alarm: a sensor judged broken by its own past or its own plausibility is not trusted even if the other sensors happen to agree with it. This conservative "weakest link" rule is what stops several jointly-failing sensors from covering for each other.

4.6 Fusion (use trusted sensors, replace untrusted ones)

An untrusted sensor is replaced by the other sensors' best guess of what it should have been, $\tilde{z}_m = \text{median}_{j \neq m} R(\bar{z}_j)$. Then all sensors are blended by their reliability:

$$z_m^* = r_m\, \bar{z}_m + (1 - r_m)\, \tilde{z}_m, \qquad \alpha_m = \frac{r_m}{\sum_k r_k}, \qquad z_{\text{fused}} = \sum_m \alpha_m\, z_m^*. \tag{8}$$

A trusted sensor is used as-is; an untrusted one is swapped for a reconstruction from the others. We then combine them with weights proportional to reliability, so trustworthy sensors count more. (If every sensor is untrusted, the weights fall back to equal.)

4.7 Module 4 — Collapse detector (is the whole system failing?)

Two whole-system warning signs are checked, using quantities we already computed:

$$c_{\text{glob}} = \begin{cases} 1 & \text{if } \overline{r^{\text{dens}}} < \tau_d \text{ (everything implausible), or} \\ 1 & \text{if cross-predictions are unusually identical on a near-constant signal,} \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

Two red flags for the system as a whole: (1) every sensor looks implausible at once (a general blackout), or (2) every sensor suddenly agrees *too* perfectly on a flat, empty-looking value—a telltale sign they all failed the same way. When either happens, the cross-checks can no longer be trusted.

4.8 Uncertainty and abstention

We turn reliability into a single confidence number and decide whether to answer:

$$U = \big(1 - \tfrac{1}{M} \textstyle\sum_m r_m\big) + \lambda\, c_{\text{glob}}, \qquad \text{abstain if } U > \tau. \tag{10}$$

Uncertainty is high when average trust in the sensors is low, or when the collapse detector fires. If uncertainty crosses a threshold, the system reports "not sure" (and can defer to a human) instead of giving a confident answer. Otherwise the prediction is $\hat{y} = h(z_{\text{fused}})$ for a simple task head h.

4.9 How it is trained (no labels saying which sensor failed)

During training we *ourselves* damage random sensors in random ways and ask the model to recover the reliability scores; the encoders and predictor R are trained with standard self-supervised objectives (matching two augmented views of the same input; matching naturally co-occurring sensors; reconstructing one sensor from the others). The per-sensor predictors T_m are trained to predict the next step on clean data, and the plausibility models p are fit on clean summaries. A small task head is trained with labels *only* for the final evaluation. Importantly, the *kinds* of damage used in training are kept separate from those used in testing, so the model learns the general idea of "unreliable," not one specific trick.

We teach the model by breaking inputs on purpose and asking it to notice, so no human has to label real failures. We then test it on *new* kinds of breakage to confirm it learned the concept, not the specific damage.

5 Architecture

5.1 (A) High-level system flow

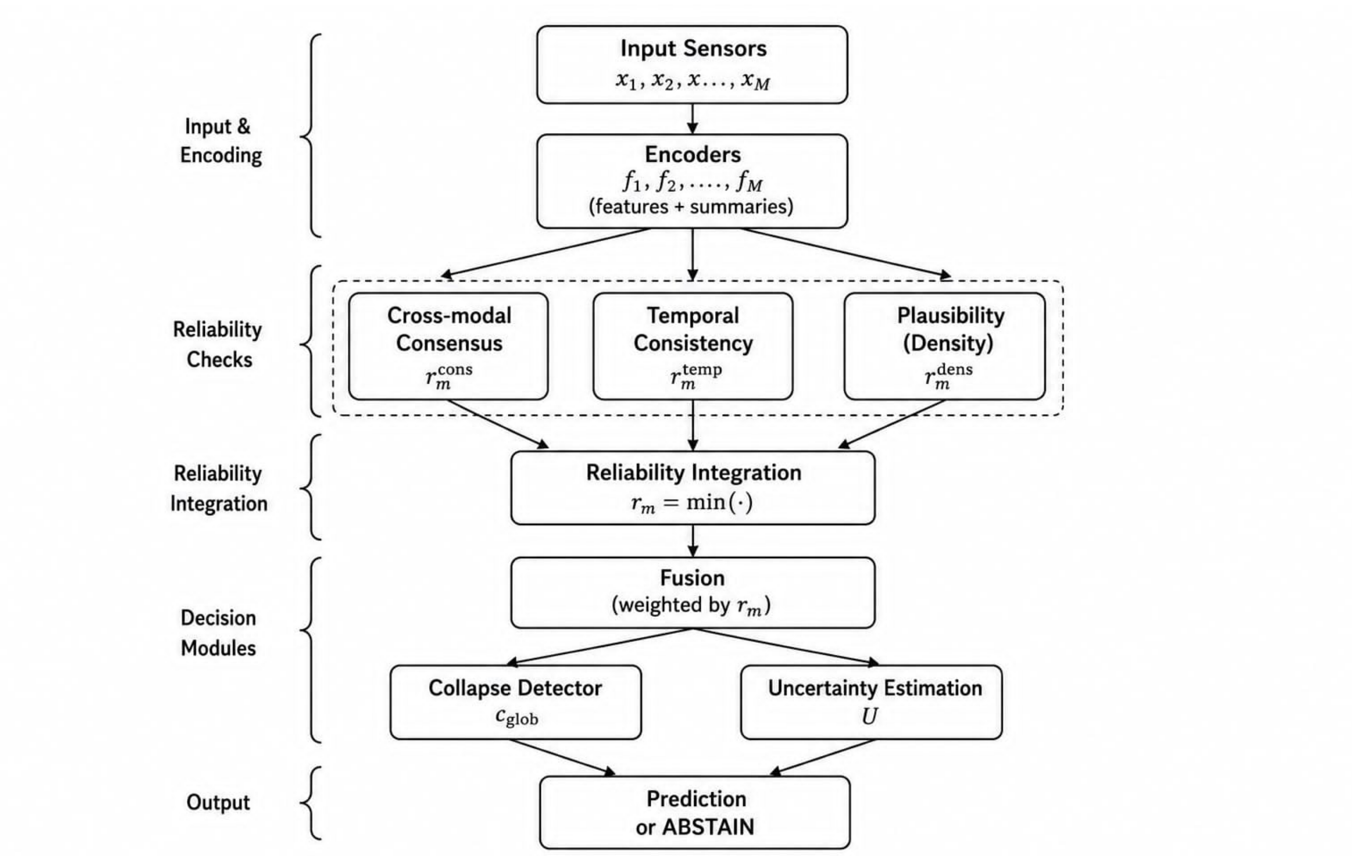

Figure 2: SIREN full reliability pipeline.

Table 1: Inputs, outputs, and roles of the SIREN modules.

Module	Input	Output	Purpose
Cross-Modal Consensus	Summary vectors from all sensors	r_m^{cons}	Measures whether sensor m agrees with the other sensors.
Temporal Consistency	Time sequence of sensor m	r_m^{temp}	Measures whether sensor m is consistent with its own recent history.
Plausibility / Density	Summary vector of sensor m	r_m^{dens}	Measures whether sensor m looks like a normal sensor reading on its own.
Reliability Integration	r_m^{cons}, r_m^{temp}, r_m^{dens}	r_m	Combines the three reliability signals using $r_m = \min(\cdot)$ to obtain the final trust score.
Fusion, Uncertainty and Collapse Detection	All reliability scores, summaries, and reconstructions	z_{fused}, U	Produces the final representation, estimates uncertainty, detects global collapse, and decides whether to abstain or predict.

5.2 (B) What each module takes in and puts out

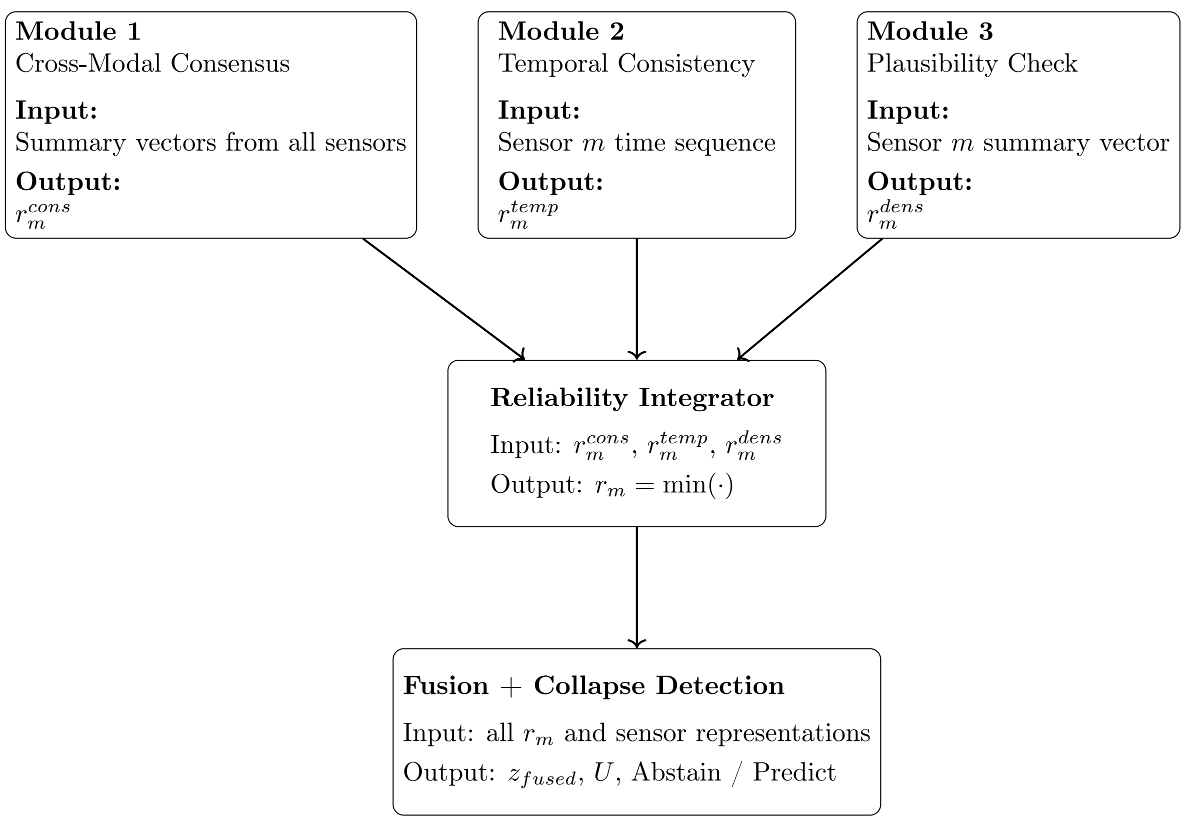

Figure 3: Inputs and outputs of each SIREN module. The three reliability signals are computed independently and combined through a conservative minimum rule before fusion and uncertainty estimation.

5.3 (C) One sample, step by step

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 1. Read each sensor over a short window; encode to sequence + summary.
 2. Module 1: predict each sensor from the others -> consensus score.
 3. Module 2: predict each sensor's present from its past -> temporal score.
 4. Module 3: score each summary against "normal" -> plausibility score.
 5. Trust r_m = the smallest of the three scores (must pass all checks).
 6. Replace low-trust sensors with a reconstruction from the others.
 7. Blend sensors with weights proportional to trust -> fused vector.
 8. Module 4: raise a global flag if the whole system looks collapsed.
 9. Compute uncertainty from low trust + global flag.
10. If uncertainty is high -> ABSTAIN; otherwise -> predict from fused vector.
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6 Limitations

We are deliberately honest about what this method cannot do.

- **Frozen vs. genuinely still.** A flat motion sensor may mean "broken" or "the person is not moving." Our checks reduce but do not fully remove this confusion.
- **A breaking point remains.** If *every* check is fooled at once (for example a broad change in conditions), the system can still be wrong; the three-check rule raises the bar, it does not make failure impossible.
- **The standalone checks must be learned well.** A poor temporal or plausibility model can cause false alarms.
- **Honest confidence under damage is hard.** Like most deep models, ours can be overconfident when inputs shift, so the uncertainty is best used for *ranking* and *abstaining*, not as an exact probability.
- **Some checks need time and structure.** The temporal check needs a time signal; plausibility is easiest for sensors with clear physical meaning.
- **Results here are simulated.** Real measurements, including real (not only synthetic) sensor faults, are required before drawing conclusions.

7 Conclusion

We described a practical and under-studied problem spotting sensors that fail quietly, with no warning flag and a simple, understandable method to handle it. SIREN judges each sensor three independent ways (agreement with others, agreement with its own past and standalone plausibility), trusts a sensor only if it passes all three, watches for whole-system collapse and abstains when it cannot tell who to trust. The components are familiar; the contribution is a clear combination that does not rely on any single check and that fails *safely.* The natural next step is to replace the illustrative results here with real measurements, including genuine hardware faults and damage types not seen during training.

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